



REF: NAB-RI-2024-09 | Issued: Q3 2024

Operation Clearwater

Transaction Fraud Detection & Risk Intelligence Report

DIVISION

Risk Intelligence

TOOL

Power BI

DATA PERIOD

Jan – Sep 2024

EST. EXPOSURE

\$2.3M+

BACKGROUND

NorthAxis Bank is a mid-sized commercial bank operating across 12 countries, serving over 80,000 customers through branch, ATM, mobile banking, and web channels. The bank processes an average of 400,000 transactions per month — spanning card payments, wire transfers, POS transactions, and online purchases.

THE INCIDENT

In Q3 2024, the bank's compliance hotline received a 340% spike in fraud-related complaints. Internal audit flagged an estimated \$2.3M in suspicious outflows concentrated in a 6-week window. The CFO has escalated the matter to the Board Risk Committee, who are convening in 72 hours. Every hour without answers increases regulatory and reputational exposure.

"We need to know who, what, when, and where — before that board meeting. Your Analysis are the only thing standing between us and a regulatory investigation."

YOUR ROLE

You have just joined NorthAxis Bank as a Junior Data Analyst on the Risk Intelligence team. You have been granted read access to the bank's Gold layer data warehouse. Using Power BI, you must investigate transaction patterns, surface anomalies, profile high-risk accounts, and deliver a data-backed risk report to the executive team.

FLAGGED SIGNALS (UNVERIFIED)

● **Velocity anomalies**

Multiple high-value transactions within minutes from the same account

● **Geographic mismatch**

Transactions from countries inconsistent with customer profiles

● **Off-hours activity**

Unusual transaction volumes between 1AM and 4AM on digital channels

● **Merchant concentration**

Disproportionate spend clustered in high-risk merchant categories

ANALYTICAL DELIVERABLES

1

Transaction Overview & Baseline KPIs

BEGINNER

Volume, value, and channel breakdown — establish what normal looks like before hunting anomalies.

2

Anomaly Detection & Velocity Checks

INTERMEDIATE

Rapid repeat transactions, off-hours spikes, and amount outliers — flag what breaks the pattern.

3

Customer Risk Profiling

INTERMEDIATE

Build behavioral baselines per customer and surface deviations from their own transaction history.

4

Merchant & Channel Risk Scoring

ADVANCED

Rank merchants and channels by their concentration of flagged and suspicious activity.

5

Fraud Risk Scoring Model

ADVANCED

Composite risk score per customer using NTILE + weighted flags — produce an actionable watchlist.

6

Executive Risk Report & Recommendations

ADVANCED

Board-ready summary — estimated exposure, top accounts to freeze, and channels to restrict.